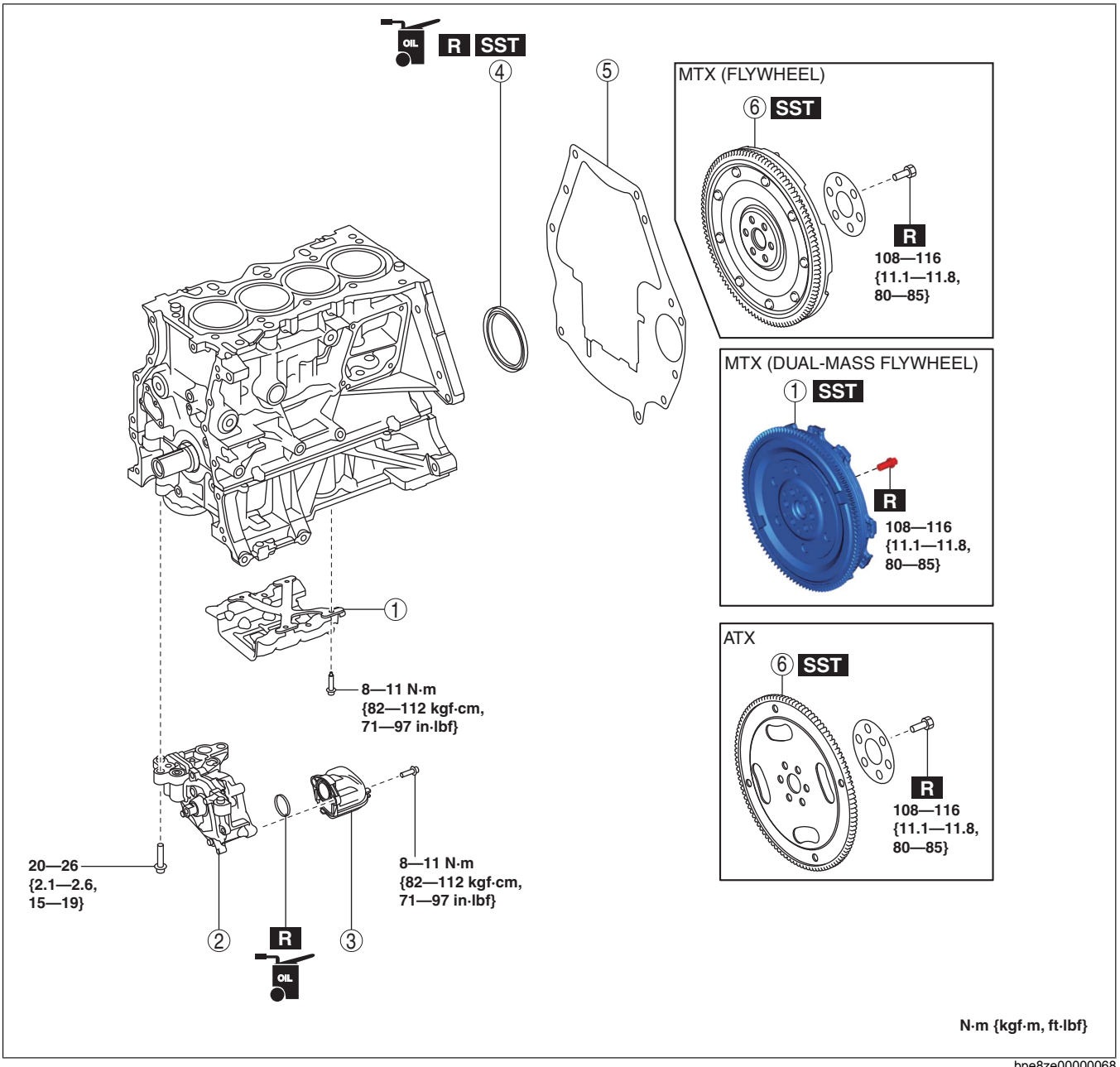
CYLINDER BLOCK ASSEMBLY (II)

id011000504100

SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0

1. Assemble in the order indicated in the table.



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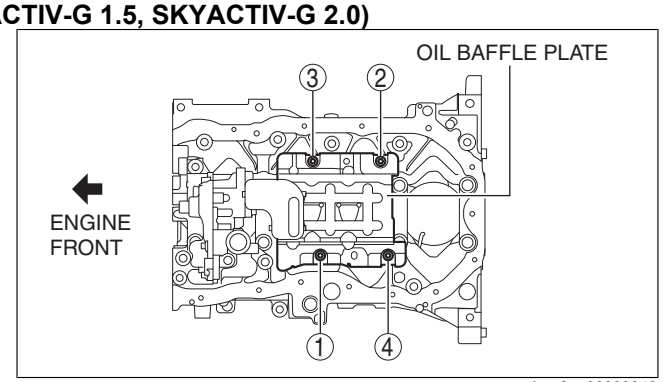
1	Oil baffle plate (See Oil baffle plate assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0).)
2	Oil pump (See Oil pump assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0).)
3	Oil strainer
4	Rear oil seal (See Rear oil seal assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0).)

5	End plate (See End plate assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0).)
6	Flywheel (MTX), drive plate (ATX) (See Flywheel (MTX), drive plate (ATX) assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0).)

Oil baffle plate assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0)

1. Tighten the bolts in the order shown in the figure.

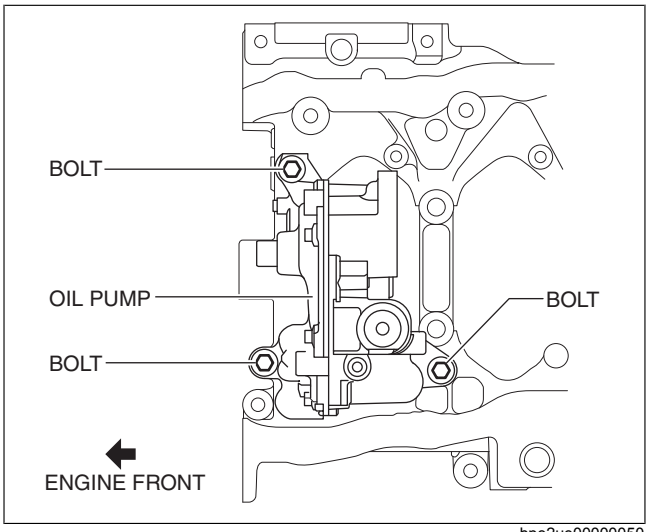
Tightening torque
8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



bpe2ue00000049

Oil pump assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0)

1. Tighten the oil pump using the following procedure:
   (1) Temporarily tighten the **three bolts** shown in the figure.

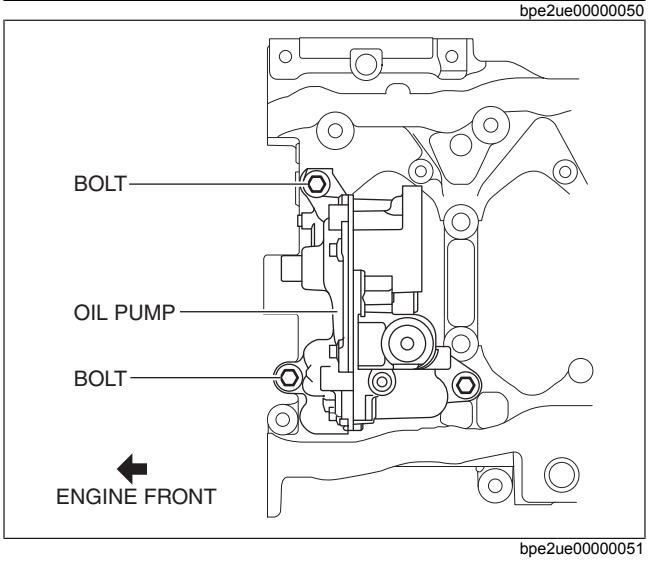


bpe2ue00000050

   (2) Tighten the **two bolts** shown in the figure to specified torque.

Note
- The tightening order for the two bolts is optional.

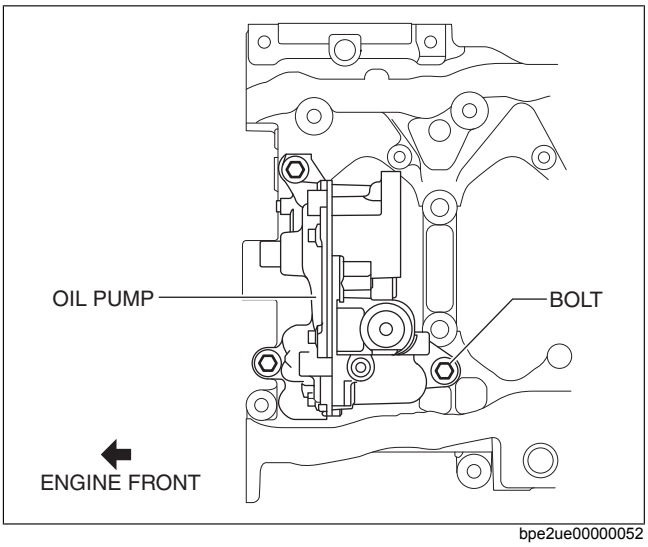
Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}



bpe2ue00000051

(3) Finally, tighten the bolt shown in the figure to the specified torque.

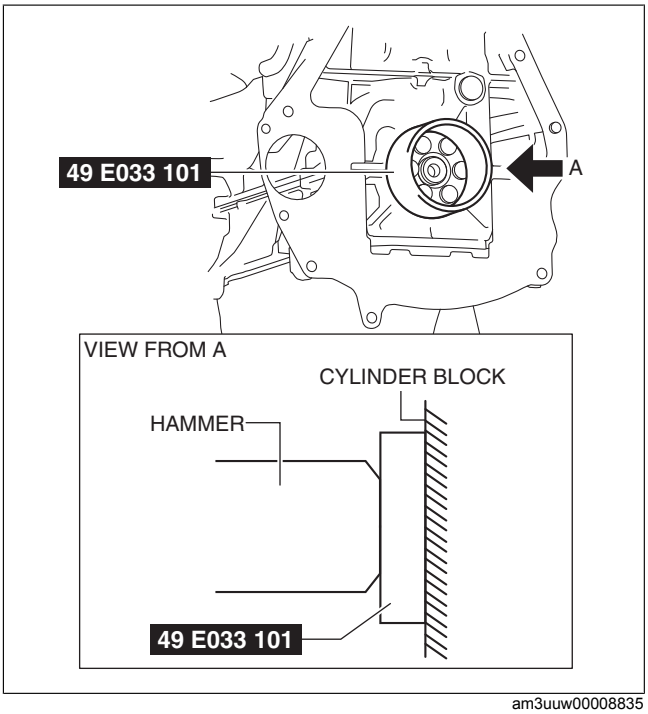
Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}



bpe2ue00000052

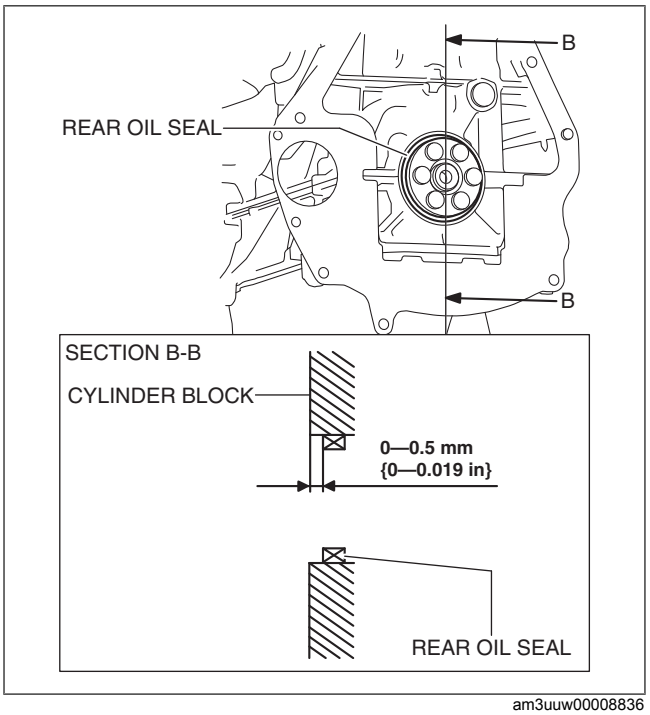
Rear oil seal assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0)

1. Apply clean engine oil to the inner surface of a new rear oil seal.
2. Insert the rear oil seal into the cylinder block by hand.
3. Tap the rear oil seal in evenly using the **SST** and a hammer.



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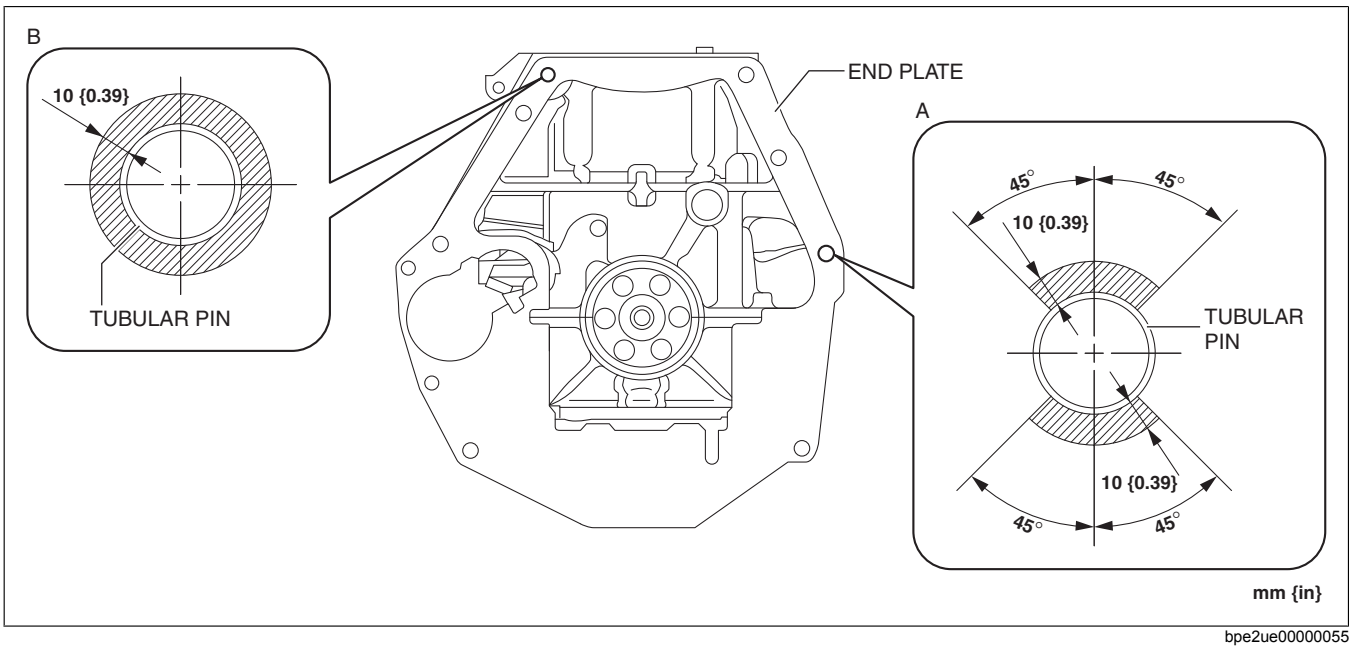
Rear oil seal press on amount
0—0.5 mm {0—0.019 in} from edge surface of cylinder block



am3uuw00008836

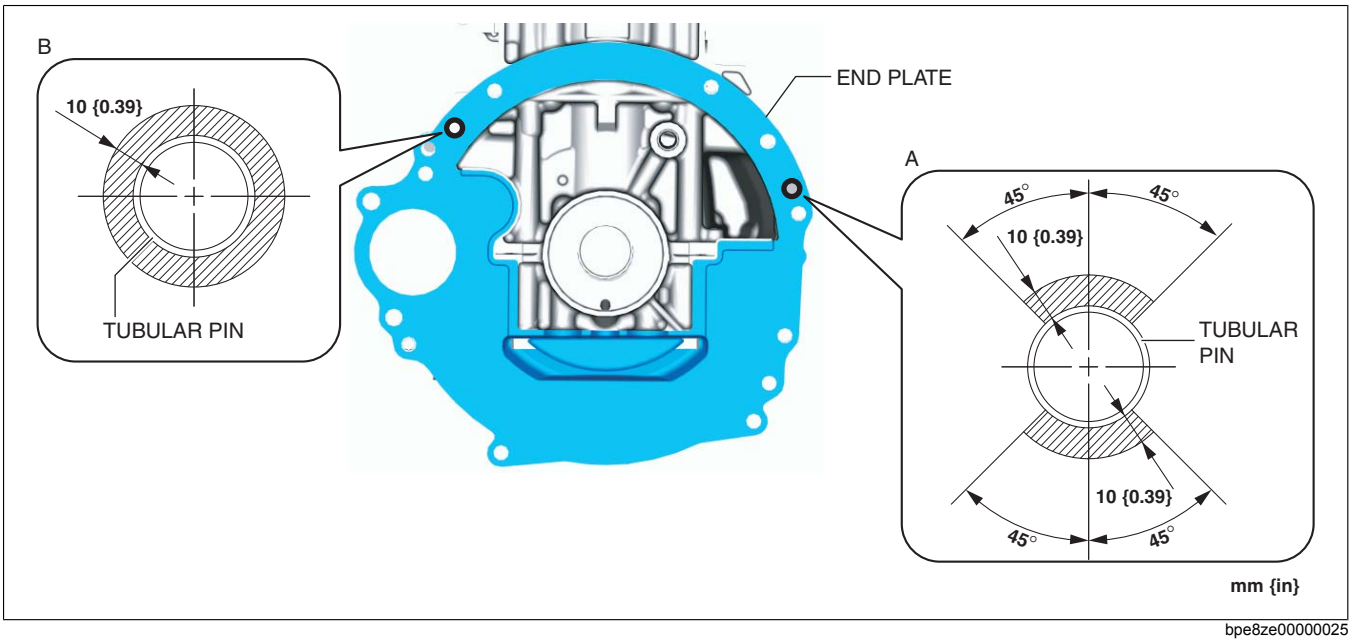
End plate assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0)
1. After end plate assembly, crimp the parts A and B shown in the figure.

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (Except MX-5 (ND)), SKYACTIV-G 2.0 (Except MX-5 (ND))



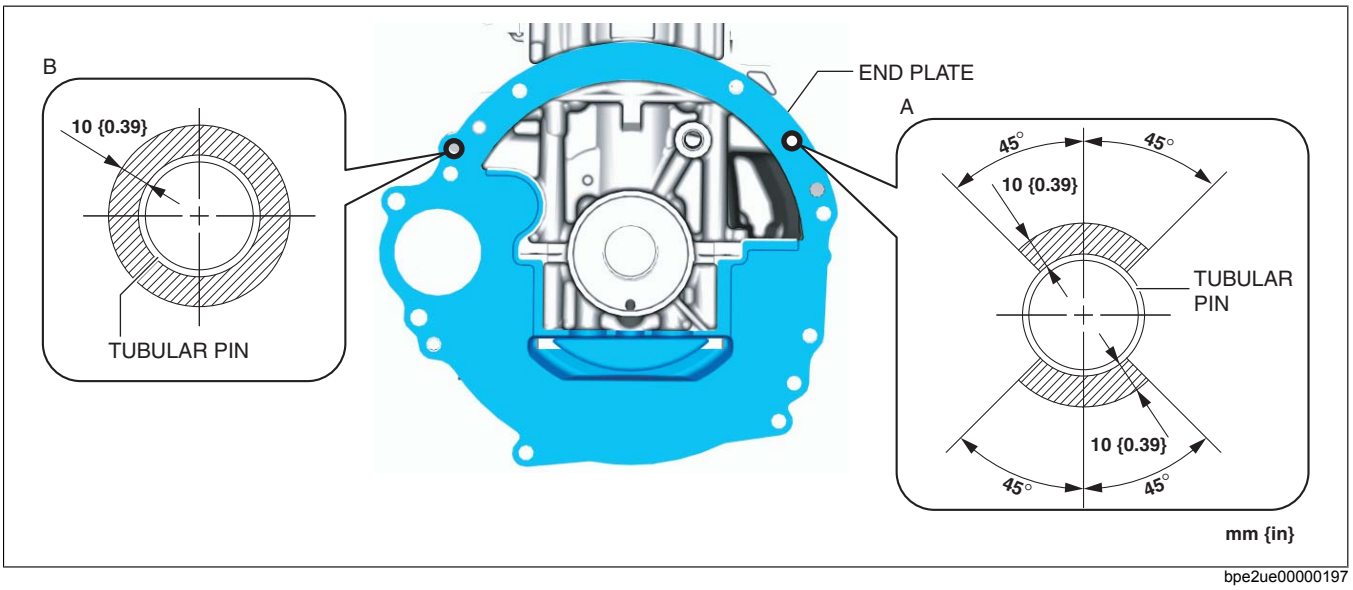
bpe2ue00000055

SKYACTIV-G 1.5 (MX-5 (ND))



bpe8ze00000025

SKYACTIV-G 2.0 (MX-5 (ND))



bpe2ue00000197

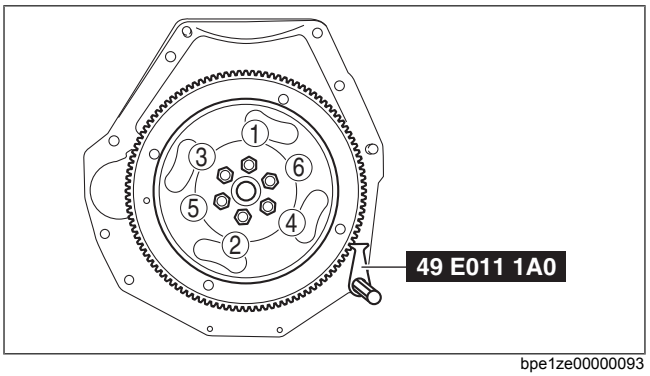
Crimp procedure
Crimp depth: 0.1—1.0 mm {0.004—0.039 in}
Crimp width: 0.5—10.0 mm {0.02—0.39 in}
Crimp locations: Part A is 1 or more on one-side within shaded area and part B is 2 or more within shaded areas

2. After crimping, verify that there is no damage and removal of the end plate.

Flywheel (MTX), drive plate (ATX) assembly note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0)
1. Hold the crankshaft using the **SST**.

2. Tighten the new installation bolts in two or three passes in the order shown in the figure.

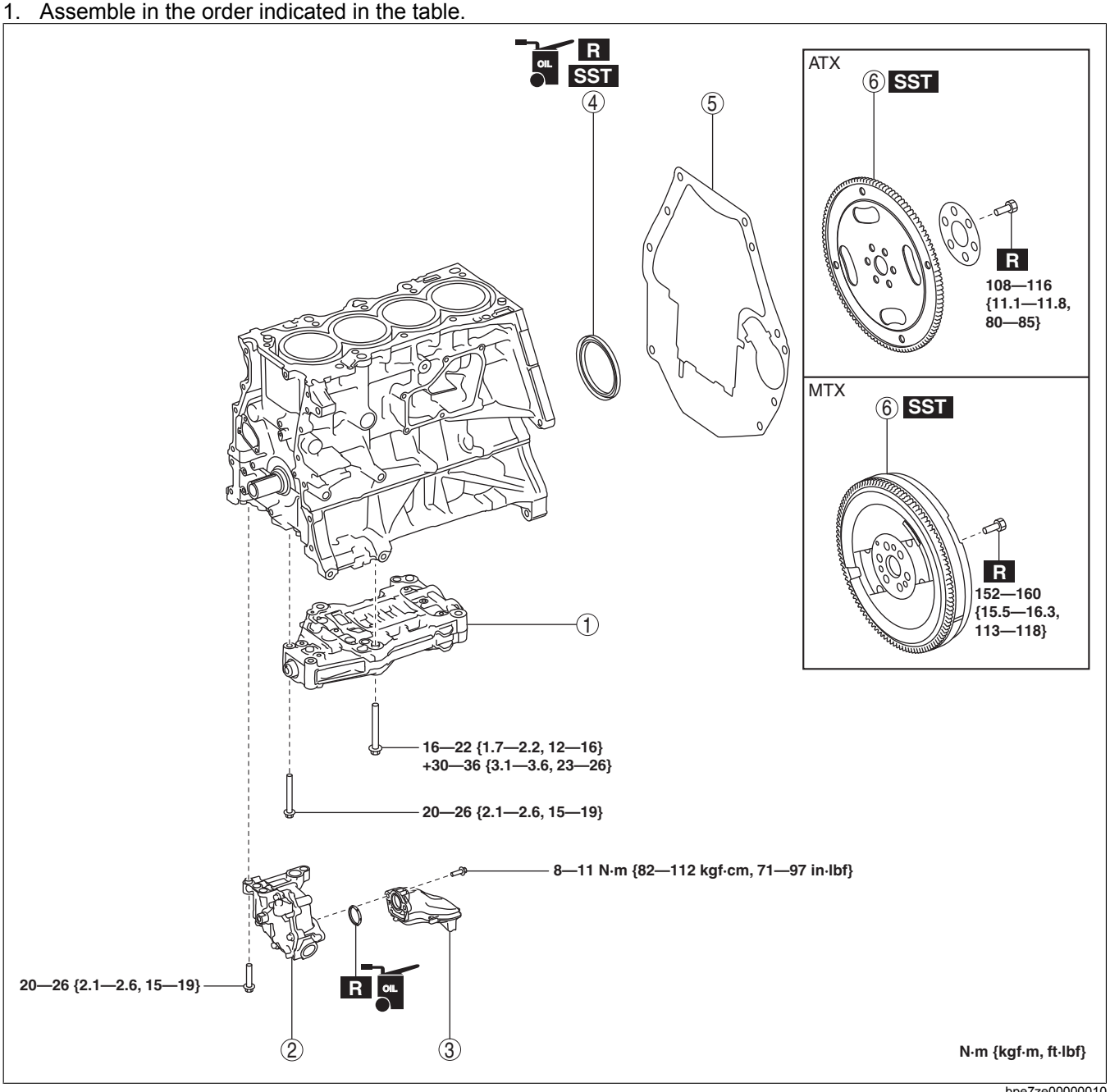
Tightening torque
108—116 N·m {11.1—11.8 kgf·m, 80—85 ft·lbf}



bpe1ze00000093

SKYACTIV-G 2.5

1. Assemble in the order indicated in the table.



bpe7ze00000010

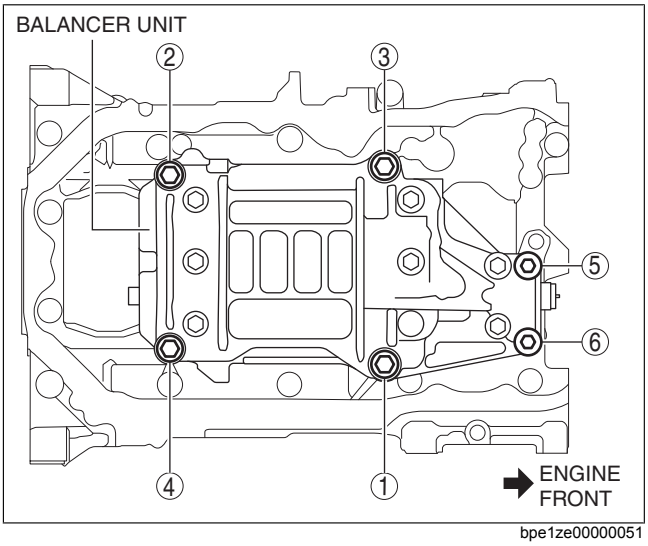
1	Balancer unit (See Balancer unit assembly note (SKYACTIV-G 2.5).)

2	Oil pump (See Oil pump assembly note (SKYACTIV-G 2.5).)
3	Oil strainer

4	Rear oil seal (See Rear oil seal assembly note (SKYACTIV-G 2.5).)
5	End plate (See End plate assembly note (SKYACTIV-G 2.5).)
6	Dual-mass flywheel (MTX), drive plate (ATX) (See Dual-mass flywheel installation bolt assembly note (SKYACTIV-G 2.5, MTX).) (See Drive plate installation bolt assembly note (SKYACTIV-G 2.5, ATX).)

Balancer unit assembly note (SKYACTIV-G 2.5)

1. Assemble the balancer unit using the following procedure:
   (1) Tighten the bolts in the order shown in the figure.



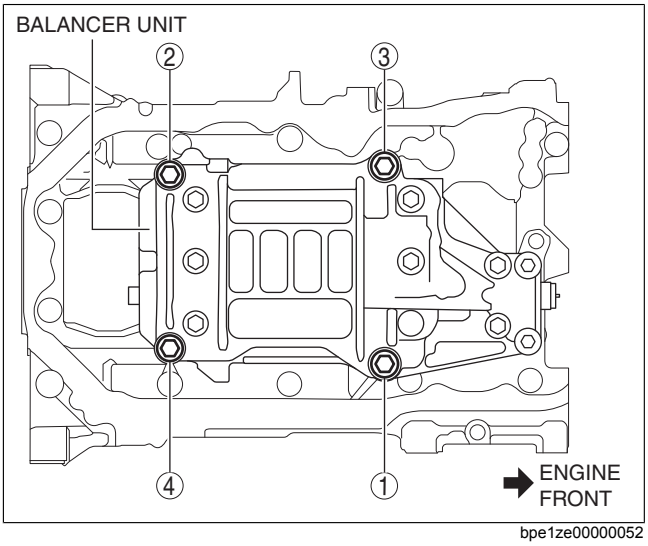
bpe1ze00000051

Tightening torque

Installation position	Tightening torque
1—4	**16—22 N·m {1.7—2.2 kgf·m, 12—16 ft·lbf}**
5, 6	**20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}**

   (2) Retighten the bolts in the order of the numbers indicated in the figure.

   Tightening torque
   30—36 N·m {3.1—3.6 kgf·m, 23—26 ft·lbf}

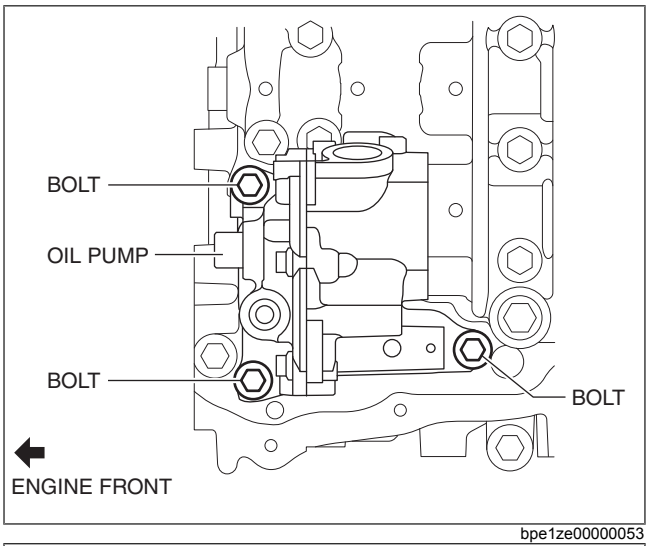


bpe1ze00000052

Oil pump assembly note (SKYACTIV-G 2.5)

1. Install the oil pump using the following procedure:

(1) Temporarily tighten the **three bolts** shown in the figure.

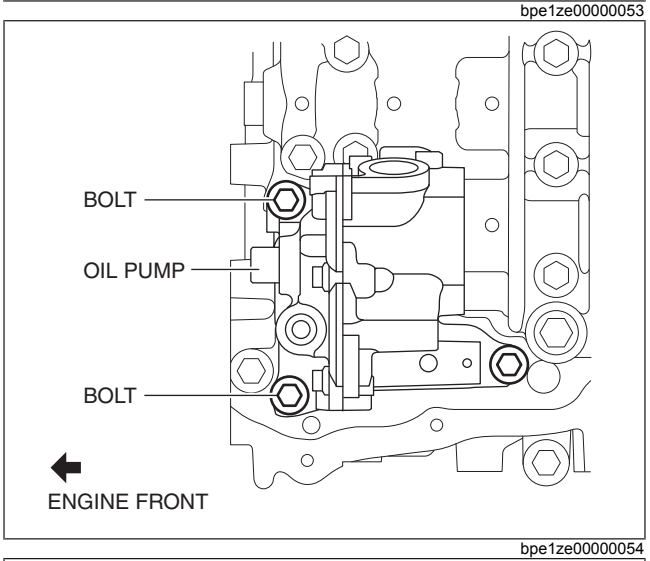


bpe1ze00000053

(2) Tighten the **two bolts** shown in the figure to the specified torque.

Note
- The tightening order for the two bolts is optional.

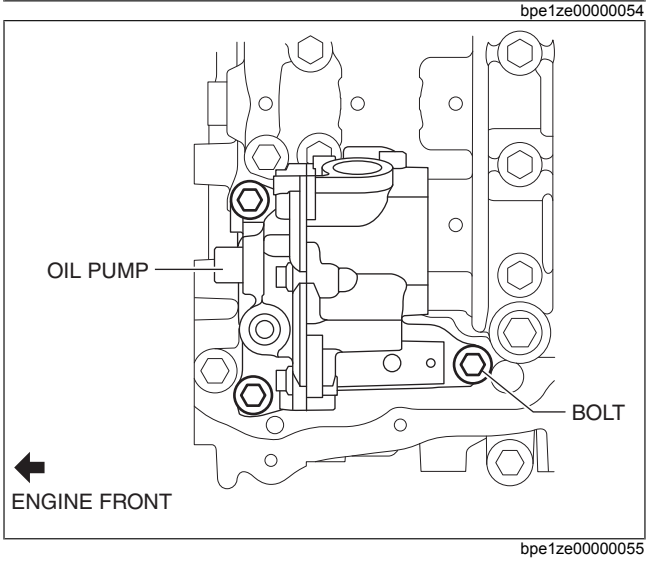
Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}



bpe1ze00000054

(3) Finally, tighten the bolt shown in the figure to the specified torque.

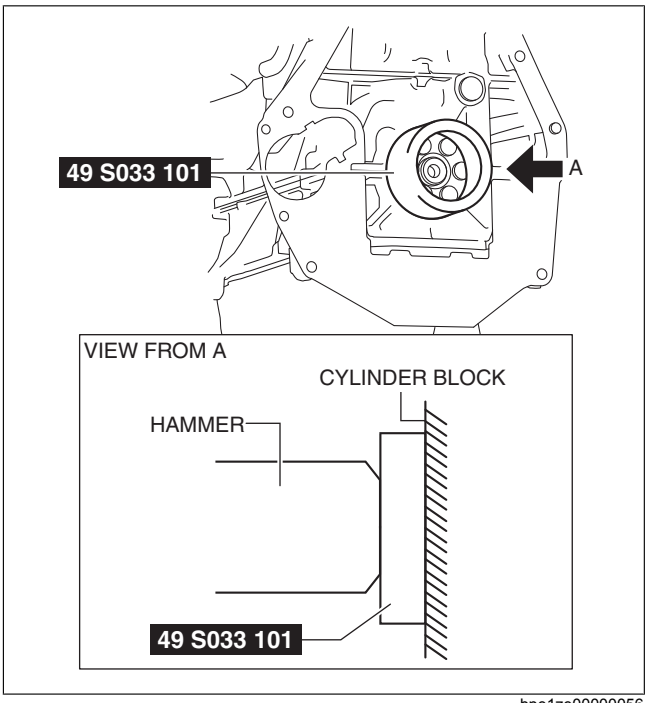
Tightening torque
20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}



bpe1ze00000055

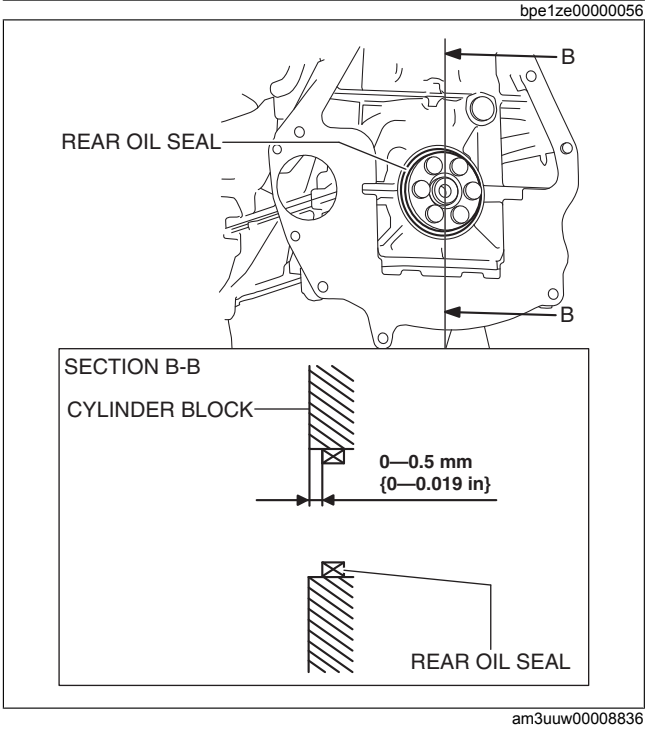
Rear oil seal assembly note (SKYACTIV-G 2.5)
1. Apply clean engine oil to the inner surface of a new rear oil seal.
2. Insert the rear oil seal into the cylinder block by hand.

3. Tap the rear oil seal in evenly using the **SST** and a hammer.



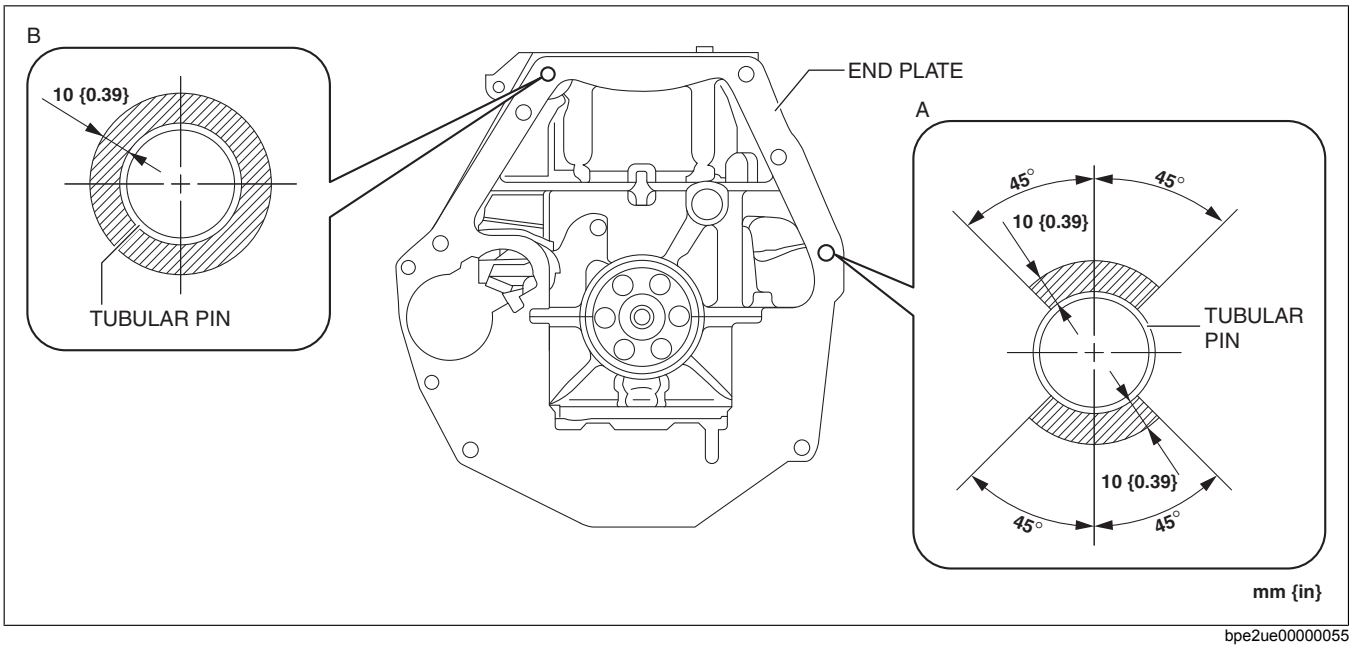
bpe1ze00000056

Rear oil seal press on amount
0—0.5 mm {0—0.019 in} from edge surface of cylinder block



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End plate assembly note (SKYACTIV-G 2.5)
1. After end plate assembly, crimp the parts A and B shown in the figure.



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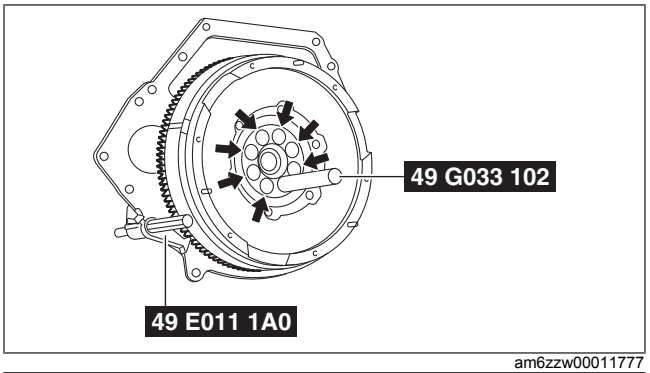
Crimp procedure
Crimp depth: 0.1—1.0 mm {0.004—0.039 in}
Crimp width: 0.5—10.0 mm {0.02—0.39 in}
Crimp locations: Part A is 1 or more on one-side within shaded area and part B is 2 or more within shaded areas

2. After crimping, verify that there is no damage and removal of the end plate.

Dual-mass flywheel installation bolt assembly note (SKYACTIV-G 2.5, MTX)
1. Hold the crankshaft using the **SST (49 E011 1A0)**.
2. Temporarily tighten the new bolts.

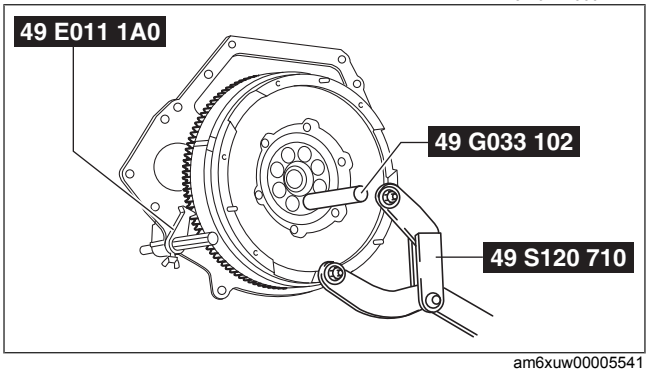
Note
- If the dual-mass flywheel (secondary flywheel side) is not positioned properly, perform the following procedure to install it:

(1) Temporarily tighten the new bolts to the position as shown in the figure.



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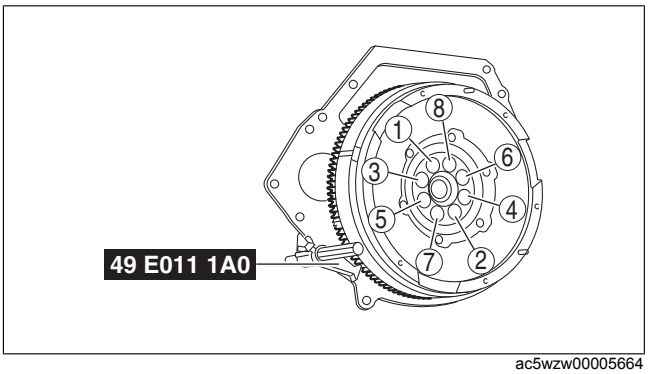
(2) Install the **SST (49 S120 710)** to the dual-mass flywheel.
(3) Rotate the dual-mass flywheel (secondary flywheel side) using the **SST (49 S120 710)**, and then temporarily tighten the remaining new bolt after removing the **SST (49 G033 102)**.



am6xuw00005541

3. Tighten the new bolts in two or three passes in the order shown in the figure.

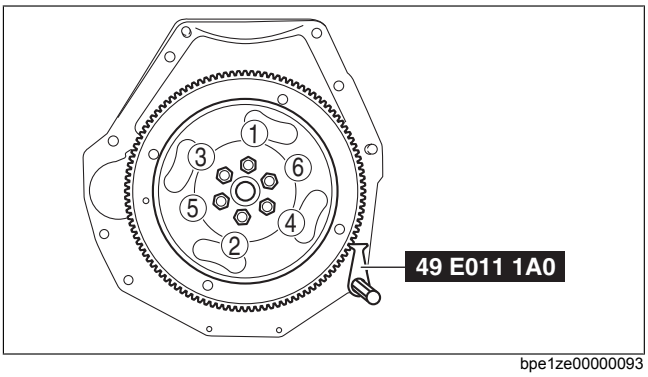
   Tightening torque
   152—160 N·m {15.5—16.3 kgf·m, 113—118 ft·lbf}



ac5wzw00005664

Drive plate installation bolt assembly note (SKYACTIV-G 2.5, ATX)
1. Hold the crankshaft using the **SST**.
2. Tighten the new bolts in two or three passes in the order shown in the figure.

   Tightening torque
   108—116 N·m {11.1—11.8 kgf·m, 80—85 ft·lbf}



bpe1ze00000093